

Q.8c (15 marks): Bio-cultural adaptation to cold environment

8. (c)

ठंडे पर्यावरण में मानव के जैव-सांस्कृतिक अनुकूलन में जनानक और अजनानक कारकों का विवेचना काजिए।

Discuss the genetic and non genetic factors in bio-cultural adaptation of human beings to cold environment.

15

इस हशिप में
नहीं लिखना
चाहिए
Candidates
must not
write on
this margin

Early Human beings evolved in the hot savanna climate of Africa. With time; they evolved bio-cultural adaptations for cold environment.

Today, humans reside at high altitudes of even 5000 metres.

BIOLOGICAL ADAPTATION

① Genetic Factors

- 1) Cold Genes found in some populations which help in adaptation to the cold

[eg] Among Brazilian highlanders

- 2) Unique Genetic Composition

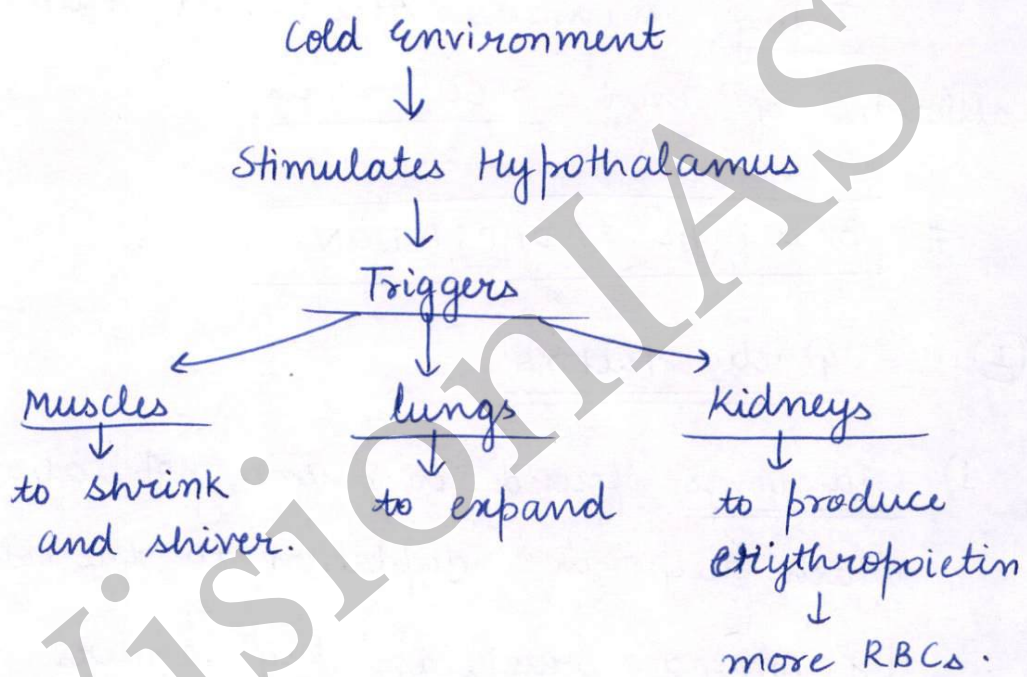
[eg] Tibetan highlanders have genes EPAS-1 and EGLN-1 which maintain Oxygen homeostasis even at high altitudes.

- 3) Enlargement of Chest Size

[eg] Peoples of Andes mountains.

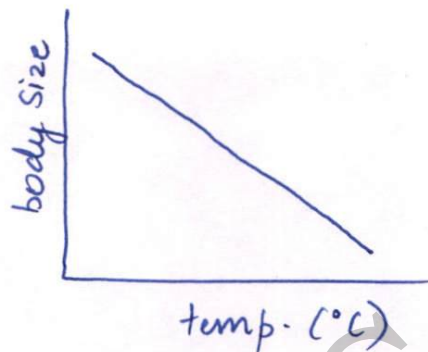
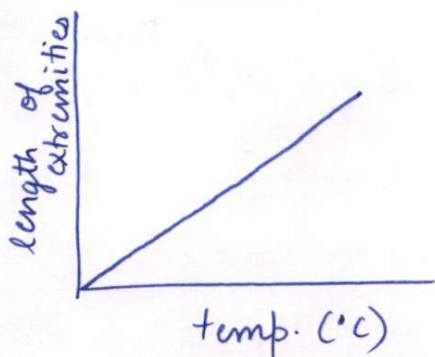
- 4) Higher levels of Haemoglobin in some populations. This helps in greater Oxygen transmission.

II Non Genetic Factors



- 1) shivering of body in response to cold climate.
- 2) Increased breathing rate to compensate for the low oxygen taken with each breath.
- 3) Kidneys increase the RBC production.

Genetic
⇒ Adaptation in Accordance with Allen
and Bergmann's Rule ÷



CULTURAL ADAPTATION

- 1) Diet → with more carbohydrates
eg Brazilian highlanders chew
herbal leaves which increase Hb.
- 2) Lifestyle → more work during the
daytime sunshine
- 3) Clothing → wooden, fur clothes.
- 4) Habitation → eg Igloos.
- 5) Modern technology → Thermostats, heaters

Thus, through the combined
effect of all these factors, human
beings are able to survive even in
harsh cold climate.